Lesson 6.2 Guided Notes

Session 1 · Dividing a Fraction by a Fraction

I can divide a fraction by another fraction using the reciprocal. · 6.NS.A.1

Word / Spanish	Meaning — complete the blank	Example
reciprocal / recíproco	A fraction flipped upside down; the two numbers multiply to make _____.	2/9 and 9/2
dividend / dividendo	The amount you _____ with and split into groups.	In 8/9 ÷ 2/9, 8/9 is the dividend
divisor / divisor	The size of each group; the number you divide _____.	In 8/9 ÷ 2/9, 2/9 is the divisor

Key idea

To divide by a fraction, multiply by its _____.

Keep the first fraction, change ÷ to ×, and flip only the _____ fraction.

Check: quotient × divisor = _____.

Worked example

How many 1/8-meter strips can be cut from 3/4 meter of tape?

Write the division: 3/4 ÷ _____.

Multiply by the reciprocal: 3/4 × _____ = 24/4.

Simplify: 24/4 = _____.

Answer: _____ strips. Check: 6 × 1/8 = 6/8 = 3/4.

You try

1. Divide.
a. 3/5 ÷ 1/5 = _____
b. 3/8 ÷ 3/4 = _____

_____

2. Ben has 9/10 pound of nuts. Each bag holds 3/10 pound. How many bags can he fill?

_____

Sum it up: To divide 2/3 ÷ 1/6, I multiply by ___ because ___.

_____

Session 2 · Dividing with Mixed Numbers

I can divide mixed numbers by writing them as improper fractions first. · 6.NS.A.1

Word / Spanish	Meaning — complete the blank	Example
mixed number / número mixto	A _____ number and a fraction written together.	2 1/2
improper fraction / fracción impropia	A fraction whose numerator is _____ than or equal to its denominator.	5/2
simplify / simplificar	Write the same amount using _____ numbers.	28/14 = 2

Key idea

Before dividing, write each mixed number as an _____ fraction.

Then multiply by the _____ of the divisor and simplify.

You cannot flip a mixed number until it is _____ fraction.

Worked example

A baker has 3 1/2 cups of flour. Each loaf uses 1 3/4 cups. How many loaves can she make?

Convert: 3 1/2 = _____ and 1 3/4 = _____.

Multiply by the reciprocal: 7/2 × _____ = 28/14.

Simplify: 28/14 = _____.

Answer: She can make _____ loaves.

You try

1. Fill in each blank.
a. 2 1/2 as an improper fraction: _____
b. 3 1/3 ÷ 2/3 = _____

_____

2. Kofi has 3 3/4 cups of rice. Each serving is 3/4 cup. How many servings can he make?

_____

Sum it up: Before I divide mixed numbers, I ___ because ___.

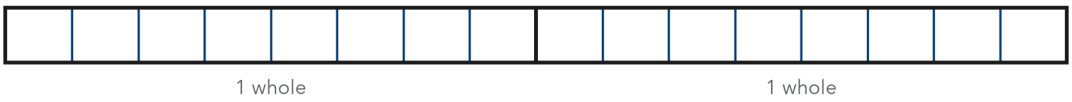
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Session 1 · Model, reason, and explain

Dividing a Fraction by a Fraction · 6.NS.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

How many 1/4-meter pieces fit in 5/8 meter? Use eighths to explain the quotient, including its fractional part.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: reports only complete pieces when the question asks for the numerical quotient.. Explain what changes and what stays true.

_____

_____

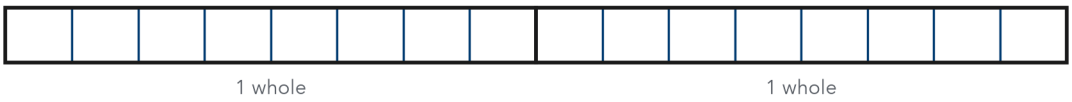
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Dividing with Mixed Numbers · 6.NS.A.1

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A roll has 3 1/2 meters. Each complete strip needs 3/4 meter. Find the numerical quotient, number of complete strips, and leftover length.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: rounds up to count a complete object that cannot be made.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?